Metric Spaces

A Four-Week Comprehensive Course Note

Metrics, Topology, Convergence, Completeness, Cantor's Theorem, Compactness, and Product Metrics

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Prepared as a four-week lecture module

Preface

This note is designed for a concentrated four-week course on metric spaces. It synthesizes the geometric and problem-oriented treatment of Kumaresan's *Topology of Metric Spaces* with the functional-analytic orientation and examples of Kreyszig's *Introductory Functional Analysis with Applications*. The purpose is not merely to list definitions, but to develop a coherent chain of ideas:

$$\text{distance} \longrightarrow \text{open sets} \longrightarrow \text{convergence and continuity} \longrightarrow \text{completeness} \longrightarrow \text{compactness.}$$

The course repeatedly uses four complementary modes of understanding:

1. geometric pictures, especially in $\mathbb{R}$, $\mathbb{R}^2$, and normed spaces;
2. abstract metric arguments based on the triangle inequality;
3. examples from sequence and function spaces;
4. counterexamples that separate notions which coincide only in special settings.

The material is organized as twelve lectures, three per week. Each chapter contains definitions, theorems with proofs, examples, pitfalls, and exercises. A final synthesis chapter and solution sketches support revision and assessment.

Course Plan

Week	Lectures	Core content
1	1–3	Course orientation; metric axioms; examples; norm-induced metrics; balls; open and closed sets.
2	4–6	Interior, closure, boundary; dense and separable spaces; sequences; continuity.
3	7–9	Cauchy sequences; completeness; Cantor's intersection theorem; compactness.
4	10–12	Product metrics; coordinatewise convergence; product completeness and compactness; synthesis.

Contents

Chapter 1

Week 1: Metric Spaces and Their Topology

Learning objectives

By the end of Week 1, students should be able to define and verify metrics, compare standard metrics, sketch metric balls, and prove the fundamental properties of open and closed sets.

1.1 Lecture 1: Course Orientation and Metric Axioms

1.1.1 Why metric spaces?

In elementary calculus, the expression $|x - y|$ measures the distance between real numbers and controls limits, continuity, and approximation. In higher-dimensional analysis, the Euclidean distance plays the same role. Functional analysis requires the same language for objects that may be functions, sequences, matrices, or solutions of differential equations. The axiomatic notion of a metric retains only the properties of distance that are essential for analysis.

Definition 1.1 (Metric space)**.** Let $X \neq \varnothing$. A function $d : X \times X \to \mathbb{R}$ is a *metric* if, for all $x, y, z \in X$,

$$\begin{aligned} &(\mathrm{M1}) \quad d(x,y) \geq 0, \\ &(\mathrm{M2}) \quad d(x,y) = 0 \iff x = y, \\ &(\mathrm{M3}) \quad d(x,y) = d(y,x), \\ &(\mathrm{M4}) \quad d(x,y) \leq d(x,z) + d(z,y). \end{aligned}$$

The pair (X, d) is called a metric space.

The elements of X are called points even when they are functions or sequences. Condition (M4) is the triangle inequality.

Proposition 1.2 (Generalized triangle inequality)**.** *For $x_1, \ldots, x_n \in X$,*

$$d(x_1, x_n) \leq \sum_{j=1}^{n-1} d(x_j, x_{j+1}).$$

Proof. Apply the triangle inequality repeatedly, or argue by induction on n. □

Proposition 1.3 (Reverse triangle inequality)**.** *For all $x, y, z \in X$,*

$$|d(x, z) - d(y, z)| \leq d(x, y).$$

More generally,

$$|d(x, y) - d(z, w)| \leq d(x, z) + d(y, w).$$

Proof. The triangle inequality gives

$$d(x, z) \leq d(x, y) + d(y, z),$$

so $d(x, z) - d(y, z) \leq d(x, y)$. Interchanging x and y gives the opposite inequality. The second statement follows by applying the reverse inequality twice. □

1.1.2 Core examples

Example 1.4 (Usual metric on $\mathbb{R}$)**.**

$$d(x, y) = |x - y|.$$

Example 1.5 (Euclidean, taxicab, and maximum metrics)**.** For $x = (x_1, \dots, x_n)$ and $y = (y_1, \dots, y_n)$ in $\mathbb{R}^n$, define

$$d_1(x, y) = \sum_{j=1}^{n} |x_j - y_j|,$$
$$d_2(x, y) = \left(\sum_{j=1}^{n} |x_j - y_j|^2 \right)^{1/2},$$
$$d_\infty(x, y) = \max_{1 \leq j \leq n} |x_j - y_j|.$$

These are respectively the taxicab, Euclidean, and maximum metrics.

Example 1.6 (Discrete metric)**.** On any nonempty set X,

$$d(x, y) = \begin{cases} 0, & x = y, \\ 1, & x \neq y. \end{cases}$$

This metric is an important source of counterexamples.

Example 1.7 (Hamming metric)**.** On the set of binary strings of length n, define $d(x, y)$ to be the number of coordinates at which x and y differ. The triangle inequality follows because every coordinate where x and z differ must be a coordinate where either x differs from y or y differs from z.

Example 1.8 (Bounded equivalent metrics)**.** If d is a metric, then

$$\rho(x, y) = \min\{1, d(x, y)\} \qquad \text{and} \qquad \sigma(x, y) = \frac{d(x, y)}{1 + d(x, y)}$$

are bounded metrics. They generate the same open sets as d.

Common misconception

A metric is not determined by the underlying set. The same set may carry many different metrics, leading to different notions of convergence, boundedness, and geometry.

1.2 Lecture 2: Normed, Sequence, and Function Spaces

Definition 1.9 (Normed linear space)**.** A norm on a real or complex vector space V is a map $\|\cdot\| : V \to \mathbb{R}$ such that

(N1) $\|x\| \geq 0$, with equality if and only if $x = 0$;

(N2) $\|\alpha x\| = |\alpha|\|x\|$;

(N3) $\|x + y\| \leq \|x\| + \|y\|$.

Proposition 1.10. *Every norm induces a metric by*

$$d(x, y) = \|x - y\|.$$

Proof. Only the triangle inequality requires comment:

$$d(x, z) = \|x - z\| = \|(x - y) + (y - z)\| \leq d(x, y) + d(y, z).$$

□

1.2.1 Sequence spaces

For $1 \leq p < \infty$,

$$\ell^p = \left\{ x = (x_n) : \sum_{n=1}^{\infty} |x_n|^p < \infty \right\}, \qquad \|x\|_p = \left(\sum_{n=1}^{\infty} |x_n|^p \right)^{1/p}.$$

For bounded sequences,

$$\ell^\infty = \{x = (x_n) : \sup_n |x_n| < \infty\}, \qquad \|x\|_\infty = \sup_n |x_n|.$$

Minkowski's inequality establishes the triangle inequality for $\|\cdot\|_p$.

1.2.2 Function spaces

Let $C[a, b]$ denote the vector space of continuous real- or complex-valued functions on $[a, b]$. Two important metrics are

$$d_\infty(f, g) = \max_{t \in [a,b]} |f(t) - g(t)|$$

and

$$d_1(f, g) = \int_a^b |f(t) - g(t)| \, dt.$$

The first controls the largest pointwise error; the second controls total accumulated error.

Key idea

In $C[a,b]$, the condition $d_\infty(f,g) < \varepsilon$ means that the graph of g lies entirely inside the vertical ε-tube surrounding the graph of f.

1.2.3 Equivalent metrics and norms

Two metrics are topologically equivalent if they generate the same open sets. On $\mathbb{R}^n$,

$$d_\infty(x,y) \le d_2(x,y) \le d_1(x,y) \le n\, d_\infty(x,y),$$

and

$$d_2(x,y) \le \sqrt{n}\, d_\infty(x,y).$$

Hence d_1, d_2, d_∞ generate the same topology.

Classroom activity

Sketch the unit balls in $\mathbb{R}^2$ for d_1, d_2, d_∞. They are respectively a diamond, a disk, and a square. Use the sketches to explain why different geometry can lead to the same open sets.

1.3 Lecture 3: Balls, Open Sets, and Closed Sets

Definition 1.11 (Balls and spheres)**.** For $x \in X$, $r > 0$, define

$$\begin{aligned} \mathrm{B}(x,r) &= \{y \in X : d(x,y) < r\}, \\ \overline{\mathrm{B}}(x,r) &= \{y \in X : d(x,y) \le r\}, \\ S(x,r) &= \{y \in X : d(x,y) = r\}. \end{aligned}$$

Definition 1.12 (Open and closed sets)**.** A set $U \subseteq X$ is open if each $x \in U$ has some $r > 0$ such that $\mathrm{B}(x,r) \subseteq U$. A set $F \subseteq X$ is closed if $X \setminus F$ is open.

Theorem 1.13. *Every open ball is open.*

Proof. Let $y \in \mathrm{B}(x,r)$ and put $s = r - d(x,y) > 0$. If $z \in \mathrm{B}(y,s)$, then

$$d(x,z) \le d(x,y) + d(y,z) < d(x,y) + s = r.$$

Thus $\mathrm{B}(y,s) \subseteq \mathrm{B}(x,r)$. □

Theorem 1.14 (Topology induced by a metric)**.** *The open subsets of a metric space satisfy:*

(T1) $\varnothing$ and X are open;

(T2) arbitrary unions of open sets are open;

(T3) finite intersections of open sets are open.

Corollary 1.15. *The closed subsets satisfy:*

(C1) $\varnothing$ and X are closed;

(C2) arbitrary intersections of closed sets are closed;

(C3) finite unions of closed sets are closed.

Example 1.16. In a discrete metric space, every singleton is open, every subset is a union of singletons, and therefore every subset is both open and closed.

Remark 1.17. A set which is both open and closed is called *clopen*. The sets $\varnothing$ and X are always clopen.

Common misconception

In a general metric space, a sphere may be empty; different radii may define the same ball; and the closure of an open ball need not equal the corresponding closed ball.

1.4 Week 1 Exercises

1.1. Determine whether $d(x, y) = |x - y|^2$ is a metric on $\mathbb{R}$.

1.2. Prove that $d(x, y) = \sqrt{|x - y|}$ is a metric on $\mathbb{R}$.

1.3. Verify that $d(x, y) = |x - y|/(1 + |x - y|)$ is a metric.

1.4. Find all balls in a discrete metric space.

1.5. Prove that every finite subset of a metric space is closed.

1.6. Show that two distinct points have disjoint open neighborhoods.

1.7. Prove that $x \mapsto d(x, a)$ is 1-Lipschitz.

1.8. Show that an arbitrary intersection of open sets need not be open.

Chapter summary

A metric provides quantitative distance and induces qualitative topology. The triangle inequality drives nearly every elementary proof. Normed spaces, sequence spaces, and function spaces illustrate why the definition is far more general than Euclidean geometry.

Chapter 2

Week 2: Closure, Density, Sequences, and Continuity

Learning objectives

By the end of Week 2, students should be able to compute closure, interior, and boundary; establish density and separability; analyze convergence; and use equivalent formulations of continuity.

2.1 Lecture 4: Interior, Closure, and Boundary

Definition 2.1 (Interior)**.** A point $x \in A$ is interior to A if some ball centered at x is contained in A. The set of interior points is A°.

Proposition 2.2. *The interior A° is the largest open subset of A. In particular,*

$$A \text{ is open} \iff A = A^\circ.$$

Definition 2.3 (Closure)**.** A point $x \in X$ belongs to $\overline{A}$ if every ball centered at x meets A:

$$x \in \overline{A} \iff \mathrm{B}(x, r) \cap A \neq \varnothing \quad \text{for all } r > 0.$$

Proposition 2.4. *The closure $\overline{A}$ is the smallest closed subset of X containing A. Thus*

$$A \text{ is closed} \iff A = \overline{A}.$$

Definition 2.5 (Accumulation and isolated points)**.** A point x is an accumulation point of A if

$$(\mathrm{B}(x, r) \setminus \{x\}) \cap A \neq \varnothing$$

for every $r > 0$. A point $x \in A$ is isolated in A if some ball centered at x meets A only at x.

If A' denotes the set of accumulation points, then

$$\overline{A} = A \cup A'.$$

Definition 2.6 (Boundary)**.** The boundary of A is

$$\partial A = \overline{A} \cap \overline{(X \setminus A)}.$$

Equivalently, every ball centered at a boundary point meets both A and its complement.

Proposition 2.7.

$$\partial A = \overline{A} \setminus A^\circ.$$

Example 2.8. For $A = (0, 1] \subset \mathbb{R}$,

$$A^\circ = (0, 1), \qquad \overline{A} = [0, 1], \qquad \partial A = \{0, 1\}.$$

For $A = \mathbb{Q} \subset \mathbb{R}$,

$$A^\circ = \varnothing, \qquad \overline{A} = \mathbb{R}, \qquad \partial A = \mathbb{R}.$$

2.2 Lecture 5: Dense and Separable Spaces

Definition 2.9 (Dense set)**.** A subset $D \subseteq X$ is dense if $\overline{D} = X$.

Theorem 2.10 (Equivalent descriptions of density)**.** *For $D \subseteq X$, the following are equivalent:*

(i) D is dense in X;

(ii) every nonempty open set meets D;

(iii) $\mathrm{B}(x, r) \cap D \neq \varnothing$ for every $x \in X$, $r > 0$;

(iv) each $x \in X$ is the limit of a sequence from D.

Definition 2.11 (Separable space)**.** A metric space is separable if it contains a countable dense subset.

Example 2.12. $\mathbb{R}$ is separable because $\mathbb{Q}$ is countable and dense. Similarly, $\mathbb{Q}^n$ is dense in $\mathbb{R}^n$, and $\mathbb{Q} + i\mathbb{Q}$ is dense in $\mathbb{C}$.

Proposition 2.13. *A discrete metric space is separable if and only if it is countable.*

Proof. In the discrete metric the only dense subset is the entire space, because $\mathrm{B}(x, 1/2) = \{x\}$. □

Theorem 2.14. *For $1 \leq p < \infty$, the space ℓ^p is separable.*

Proof. Consider sequences with finite support and rational coordinates. This set is countable. Given $x \in \ell^p$, first truncate its tail so that the tail has small p-norm, and then approximate its finitely many remaining coordinates by rationals. □

Theorem 2.15. *The space ℓ^∞ is not separable.*

Proof. There are uncountably many binary sequences. Any two distinct binary sequences have ℓ^∞-distance 1. Therefore, the balls of radius $1/3$ centered at binary sequences are pairwise disjoint. A dense set must meet every one of them, so it cannot be countable. □

Theorem 2.16. *Every separable metric space has a countable base.*

Proof. If $D = \{d_1, d_2, \dots\}$ is countable and dense, then

$$\mathcal{B} = \{\mathrm{B}(d_j, 1/m) : j, m \in \mathbb{N}\}$$

is a countable base. □

2.3 Lecture 6: Sequences and Continuity

Definition 2.17 (Convergence)**.** A sequence (x_n) converges to $x \in X$, written $x_n \to x$, if

$$\forall \varepsilon > 0 \ \exists N \ \forall n \geq N : \quad d(x_n, x) < \varepsilon.$$

Theorem 2.18. *The limit of a convergent sequence in a metric space is unique.*

Proof. If $x_n \to x$ and $x_n \to y$, then

$$d(x, y) \leq d(x, x_n) + d(x_n, y) \to 0.$$

Thus $d(x, y) = 0$, so $x = y$. □

Proposition 2.19. *Every convergent sequence is bounded, and every subsequence of a convergent sequence converges to the same limit.*

Theorem 2.20 (Sequential characterization of closure)**.**

$$x \in \overline{A} \iff \textit{there exists } (a_n) \subseteq A \textit{ with } a_n \to x.$$

Proof. If $x \in \overline{A}$, choose $a_n \in A \cap \mathrm{B}(x, 1/n)$. Conversely, if $a_n \to x$, every ball about x contains all sufficiently late terms. □

Corollary 2.21 (Sequential characterization of closed sets)**.** *A subset F is closed if and only if every convergent sequence in F has its limit in F.*

Definition 2.22 (Continuity)**.** A map $f : (X, d_X) \to (Y, d_Y)$ is continuous at $x \in X$ if, for every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$d_X(x, y) < \delta \implies d_Y(f(x), f(y)) < \varepsilon.$$

Theorem 2.23 (Sequential criterion)**.** *A map $f : X \to Y$ is continuous at x if and only if*

$$x_n \to x \implies f(x_n) \to f(x).$$

Proof. The forward direction follows immediately from the ε-δ definition. For the converse, if continuity fails, there is an $\varepsilon_0 > 0$ such that for each n one can choose x_n with $d_X(x_n, x) < 1/n$ but $d_Y(f(x_n), f(x)) \geq \varepsilon_0$. Then $x_n \to x$ while $f(x_n) \not\to f(x)$. □

Theorem 2.24 (Open-set criterion)**.** *A map $f : X \to Y$ is continuous if and only if $f^{-1}(U)$ is open in X for every open set $U \subseteq Y$. Equivalently, inverse images of closed sets are closed.*

Definition 2.25 (Uniform and Lipschitz continuity)**.** The map $f : X \to Y$ is uniformly continuous if, for every $\varepsilon > 0$, there is a $\delta > 0$, independent of the point, such that

$$d_X(x, y) < \delta \implies d_Y(f(x), f(y)) < \varepsilon.$$

It is Lipschitz if there exists $L \geq 0$ such that

$$d_Y(f(x), f(y)) \leq L d_X(x, y).$$

Every Lipschitz map is uniformly continuous.

Example 2.26 (Distance to a set)**.** For nonempty $A \subseteq X$, define

$$d(x, A) = \inf_{a \in A} d(x, a).$$

Then

$$|d(x, A) - d(y, A)| \leq d(x, y),$$

so the distance-to-set function is 1-Lipschitz.

2.4 Week 2 Exercises

2.1. Find interior, closure, accumulation points, and boundary for $(0, 1]$, $\mathbb{Q}$, $\mathbb{Z}$, and $\{1/n : n \in \mathbb{N}\}$.

2.2. Prove $\overline{(A \cup B)} = \overline{A} \cup \overline{B}$.

2.3. Determine whether $\overline{(A \cap B)} = \overline{A} \cap \overline{B}$ always holds.

2.4. Prove that a continuous surjection maps dense sets to dense sets.

2.5. If continuous maps $f, g : X \to Y$ agree on a dense subset and Y is metric, prove $f = g$.

2.6. Characterize convergent sequences in a discrete metric space.

2.7. Prove the open-set criterion for continuity.

2.8. Give a continuous function which is not uniformly continuous.

Chapter summary

Closure and density translate arbitrary approximation into the language of balls and sequences. In metric spaces, sequential methods are especially powerful: they characterize closure, closedness, and continuity.

Chapter 3

Week 3: Completeness, Cantor's Theorem, and Compactness

Learning objectives

By the end of Week 3, students should be able to analyze Cauchy sequences, prove completeness results, apply Cantor's intersection theorem, and use open-cover and sequential compactness.

3.1 Lecture 7: Cauchy Sequences and Completeness

Definition 3.1 (Cauchy sequence)**.** A sequence (x_n) is Cauchy if

$$\forall \varepsilon > 0 \ \exists N \ \forall m, n \geq N : \quad d(x_m, x_n) < \varepsilon.$$

Theorem 3.2. *Every convergent sequence is Cauchy.*

Proof. If $x_n \to x$, then for sufficiently large m, n,

$$d(x_m, x_n) \leq d(x_m, x) + d(x, x_n) < \varepsilon.$$

□

Proposition 3.3. *Every Cauchy sequence is bounded.*

Definition 3.4 (Complete metric space)**.** A metric space is complete if every Cauchy sequence converges to a point of the space.

Example 3.5. The spaces $\mathbb{R}$, $\mathbb{C}$, $\mathbb{R}^n$, $\mathbb{C}^n$, ℓ^p for $1 \leq p \leq \infty$, and $C[a, b]$ under the supremum metric are complete. The spaces $\mathbb{Q}$ and $(0, 1)$ under their usual metrics are not complete.

Theorem 3.6 (Closed subspaces)**.** *A closed subspace of a complete metric space is complete.*

Proof. A Cauchy sequence in the subspace is Cauchy in the ambient space and hence converges there. Closedness forces the limit to remain in the subspace. □

Theorem 3.7 (Complete subspaces are closed)**.** *Every complete subspace of a metric space is closed.*

Proof. If $x \in \overline{A}$, choose $a_n \in A$ with $a_n \to x$. The sequence is Cauchy in A, and completeness gives a limit $a \in A$. Uniqueness of limits gives $a = x$. □

Corollary 3.8. *If X is complete, then a subspace $A \subseteq X$ is complete if and only if it is closed.*

Theorem 3.9. *A Cauchy sequence with a convergent subsequence converges to the same limit.*

Proof. Suppose $x_{n_k} \to x$. Given $\varepsilon > 0$, choose N so that $d(x_m, x_n) < \varepsilon/2$ for $m, n \geq N$. Choose k with $n_k \geq N$ and $d(x_{n_k}, x) < \varepsilon/2$. Then, for $n \geq N$,

$$d(x_n, x) \leq d(x_n, x_{n_k}) + d(x_{n_k}, x) < \varepsilon.$$

□

3.1.1 Completeness of $C[a, b]$ in the sup norm

Let (f_n) be Cauchy under $\|\cdot\|_\infty$. For each fixed t, $(f_n(t))$ is Cauchy in $\mathbb{R}$ or $\mathbb{C}$, so define $f(t) = \lim_n f_n(t)$. The Cauchy estimate is uniform in t, hence $f_n \to f$ uniformly. A uniform limit of continuous functions is continuous. Therefore $C[a, b]$ is complete.

3.2 Lecture 8: Cantor's Intersection Theorem

Definition 3.10 (Diameter)**.** For nonempty $A \subseteq X$,

$$\operatorname{diam}(A) = \sup\{d(x, y) : x, y \in A\}.$$

Theorem 3.11 (Cantor intersection theorem)**.** *Let (X, d) be complete. Suppose*

$$F_1 \supseteq F_2 \supseteq F_3 \supseteq \cdots$$

is a nested sequence of nonempty closed sets such that $\operatorname{diam}(F_n) \to 0$. *Then*

$$\bigcap_{n=1}^{\infty} F_n$$

consists of exactly one point.

Proof. Choose $x_n \in F_n$. Given $\varepsilon > 0$, select N such that $\operatorname{diam}(F_N) < \varepsilon$. If $m, n \geq N$, nestedness gives $x_m, x_n \in F_N$, so

$$d(x_m, x_n) \leq \operatorname{diam}(F_N) < \varepsilon.$$

Thus (x_n) is Cauchy and converges to some $x \in X$. For fixed k, all x_n with $n \geq k$ lie in F_k. Since F_k is closed, $x \in F_k$. Hence x belongs to the intersection. If x, y both belong to the intersection, then

$$d(x, y) \leq \operatorname{diam}(F_n)$$

for every n; therefore $d(x, y) = 0$. □

Common misconception

All three hypotheses matter. Without completeness, the limiting point may be missing; without closedness, the limit may lie outside every set; without shrinking diameter, the intersection may be empty or contain many points.

Theorem 3.12 (Converse characterization)**.** *A metric space is complete if every nested sequence of nonempty closed sets whose diameters tend to zero has nonempty intersection.*

3.3 Lecture 9: Compactness

Definition 3.13 (Compactness)**.** A subset $K \subseteq X$ is compact if every open cover of K has a finite subcover.

Theorem 3.14. *Every compact subset of a metric space is bounded.*

Proof. Fix $x_0 \in X$. The balls $\mathrm{B}(x_0, n)$, $n \in \mathbb{N}$, cover K. A finite subcover exists, so one sufficiently large ball contains K. $\square$

Theorem 3.15. *Every compact subset of a metric space is closed.*

Theorem 3.16. *A closed subset of a compact space is compact.*

Definition 3.17 (Sequential compactness)**.** A metric space is sequentially compact if every sequence has a convergent subsequence with limit in the space.

Theorem 3.18 (Metric compactness equivalences)**.** *For a metric space X, the following are equivalent:*

- *(i) X is compact;*
- *(ii) X is sequentially compact;*
- *(iii) every infinite subset of X has an accumulation point;*
- *(iv) X is complete and totally bounded.*

Definition 3.19 (Total boundedness)**.** A metric space X is totally bounded if, for every $\varepsilon > 0$, there exist finitely many points $x_1, \ldots, x_N$ such that

$$X \subseteq \bigcup_{j=1}^{N} \mathrm{B}(x_j, \varepsilon).$$

Theorem 3.20 (Heine–Borel)**.** *A subset of $\mathbb{R}^n$ is compact if and only if it is closed and bounded.*

Common misconception

Heine–Borel is special to finite-dimensional Euclidean spaces. In arbitrary metric spaces, closed and bounded does not imply compact. An infinite discrete metric space is the standard counterexample.

Theorem 3.21 (Continuous image)**.** *If K is compact and $f : K \to Y$ is continuous, then $f(K)$ is compact.*

Corollary 3.22 (Extreme value theorem)**.** *A continuous real-valued function on a compact metric space is bounded and attains its maximum and minimum.*

Theorem 3.23 (Heine–Cantor)**.** *Every continuous map from a compact metric space into a metric space is uniformly continuous.*

Lemma 3.24 (Lebesgue number lemma)**.** *If K is compact and $\mathcal{U}$ is an open cover of K, then there exists $\delta > 0$ such that every subset of K of diameter less than δ lies in some member of $\mathcal{U}$.*

3.4 Week 3 Exercises

3.1. Show that $1/n$ is Cauchy in $(0, 1)$ but does not converge there.

3.2. Prove that compact metric spaces are complete.

3.3. Give a bounded metric space that is not totally bounded.

3.4. Prove Cantor's intersection theorem and construct counterexamples when each hypothesis is removed.

3.5. Determine whether $(0, 1)$, $[0, 1]$, $\{1/n : n \in \mathbb{N}\}$, and $\{0\} \cup \{1/n : n \in \mathbb{N}\}$ are compact.

3.6. Prove that the continuous image of a compact set is compact.

3.7. Prove the extreme value theorem.

3.8. Show that a compact metric space is separable.

Chapter summary

Completeness controls Cauchy sequences; total boundedness controls the global spread of a space; compactness combines them. Cantor's theorem converts nested geometric information into existence and uniqueness of a point.

Chapter 4

Week 4: Product Metrics and Structural Synthesis

Learning objectives

By the end of Week 4, students should be able to work with standard product metrics and transfer convergence, continuity, completeness, compactness, density, and separability between products and their factors.

4.1 Lecture 10: Construction and Equivalence of Product Metrics

Let (X, d_X) and (Y, d_Y) be metric spaces. For $p = (x_1, y_1)$ and $q = (x_2, y_2)$, define

$$\begin{aligned} d_\infty(p, q) &= \max\{d_X(x_1, x_2), d_Y(y_1, y_2)\}, \\ d_1(p, q) &= d_X(x_1, x_2) + d_Y(y_1, y_2), \\ d_2(p, q) &= \left(d_X(x_1, x_2)^2 + d_Y(y_1, y_2)^2\right)^{1/2}. \end{aligned}$$

Each is a metric on $X \times Y$.

Proposition 4.1. *The three product metrics above are topologically equivalent.*

Proof. For $a, b \geq 0$,

$$\max\{a, b\} \leq \sqrt{a^2 + b^2} \leq a + b \leq 2 \max\{a, b\}.$$

Apply this with $a = d_X(x_1, x_2)$, $b = d_Y(y_1, y_2)$. □

Proposition 4.2 (Balls in the maximum metric)**.**

$$\mathrm{B}_{d_\infty}((x, y), r) = \mathrm{B}_X(x, r) \times \mathrm{B}_Y(y, r).$$

Proposition 4.3. *If $U \subseteq X$ and $V \subseteq Y$ are open, then $U \times V$ is open in $X \times Y$. The coordinate projections*

$$\pi_X(x, y) = x, \qquad \pi_Y(x, y) = y$$

are continuous and open.

4.2 Lecture 11: Convergence, Continuity, Completeness, and Compactness in Products

Theorem 4.4 (Coordinatewise convergence). *For any standard finite product metric,*

$$(x_n, y_n) \to (x, y) \iff x_n \to x \text{ in } X \text{ and } y_n \to y \text{ in } Y.$$

Theorem 4.5 (Maps into products). *A map*

$$F : Z \to X \times Y, \qquad F(z) = (f(z), g(z)),$$

is continuous if and only if both component maps $f : Z \to X$ and $g : Z \to Y$ are continuous.

Theorem 4.6 (Product completeness). *If X and Y are complete, then $X \times Y$, equipped with a standard product metric, is complete.*

Proof. A Cauchy sequence (x_n, y_n) has Cauchy coordinate sequences. Let $x_n \to x$ and $y_n \to y$. Coordinatewise convergence gives $(x_n, y_n) \to (x, y)$. □

Theorem 4.7 (Product compactness). *If X and Y are compact metric spaces, then $X \times Y$ is compact.*

Proof. Given a sequence (x_n, y_n), choose a subsequence (x_{n_k}) converging in X. From the corresponding (y_{n_k}), choose a further convergent subsequence. The paired subsequence converges coordinatewise. Hence the product is sequentially compact and therefore compact. □

Proposition 4.8 (Products and separability). *If X and Y are separable, then $X \times Y$ is separable.*

Proof. If D_X and D_Y are countable dense subsets, then $D_X \times D_Y$ is countable and dense in the product. □

4.3 Lecture 12: Synthesis of the Main Concepts

4.3.1 Implication diagram

For metric spaces,

$$\boxed{\text{compact}} \implies \boxed{\text{complete}}$$

and

$$\boxed{\text{compact}} \implies \boxed{\text{totally bounded}} \implies \boxed{\text{bounded}}.$$

Moreover,

$$\boxed{\text{complete} + \text{totally bounded}} \iff \boxed{\text{compact}}.$$

4.3.2 Counterexample bank

Complete but not compact:
$\mathbb{R}$ with the usual metric.

Totally bounded but not complete:
$(0, 1)$ with the usual metric.

Bounded but not totally bounded:
an infinite discrete metric space.

Closed and bounded but not compact:
an infinite discrete metric space in itself.

Infinite compact set:
$\{0\} \cup \{1/n : n \in \mathbb{N}\} \subset \mathbb{R}$.

4.3.3 A useful interpretation

- Completeness prevents Cauchy sequences from converging to missing points.
- Total boundedness prevents infinitely many points from remaining uniformly separated.
- Compactness combines both controls.
- Density concerns approximation by a subset; separability asks whether a countable subset suffices.
- Equivalent metrics preserve topology but need not preserve numerical distances or boundedness.

Theorem 4.9. *Every compact metric space is separable.*

Proof. For each n, total boundedness yields a finite $1/n$-net F_n. Then $D = \bigcup_n F_n$ is countable. Given $x \in X$ and $r > 0$, choose n with $1/n < r$; some point of F_n lies within $1/n$ of x. Thus D is dense. □

4.4 Week 4 Exercises

4.1. Verify the metric axioms for d_∞, d_1, and d_2 on $X \times Y$.

4.2. Prove that these product metrics are equivalent.

4.3. Prove coordinatewise convergence.

4.4. Prove that the coordinate projections are 1-Lipschitz for the maximum metric.

4.5. Prove product completeness.

4.6. Prove product compactness using sequential compactness.

4.7. Prove that the product of separable metric spaces is separable.

4.8. Show that $A \times B$ is closed in $X \times Y$ whenever A and B are closed.

Chapter summary

Standard finite product metrics are equivalent and encode coordinatewise behavior. Completeness, compactness, and separability pass naturally to finite products. Product spaces therefore provide an efficient setting for systems with several interacting components.

Chapter 5

Integrated Problems and Assessment

5.1 Conceptual Questions

1. Why is the triangle inequality central to the topology of metric spaces?
2. In what sense does a metric contain more information than the topology it induces?
3. Why is separability an approximation property rather than a cardinality property?
4. Explain the difference between Cauchy convergence and ordinary convergence.
5. Why is total boundedness stronger than boundedness?
6. Which features of compact subsets of $\mathbb{R}^n$ remain true in arbitrary metric spaces?
7. Why does the sequential characterization of continuity work particularly well in metric spaces?

5.2 Proof Problems

1. Prove that $x \mapsto d(x, A)$ is continuous for every nonempty $A \subseteq X$.
2. Prove that a compact metric space is complete.
3. Prove that a complete and totally bounded metric space is compact.
4. Prove that a continuous map from a compact metric space is uniformly continuous.
5. Prove that a compact metric space has a countable dense subset.
6. Prove that if D is dense and $f, g : X \to Y$ are continuous and agree on D, then they agree everywhere.
7. Prove that every isometry is continuous and maps Cauchy sequences to Cauchy sequences.

5.3 Counterexample Problems

Construct examples showing that:

1. a closed and bounded metric space need not be compact;

2. a Cauchy sequence need not converge in an incomplete space;
3. a continuous function need not be uniformly continuous;
4. an infinite intersection of open sets need not be open;
5. the closure of an open ball need not equal the corresponding closed ball;
6. a bounded metric space need not be totally bounded;
7. equivalent metrics need not assign the same numerical distances.

5.4 Capstone Application: Comparing Metrics on $C[0, 1]$

Consider

$$d_\infty(f, g) = \max_{t\in[0,1]} |f(t) - g(t)|$$

and

$$d_1(f, g) = \int_0^1 |f(t) - g(t)|\, dt.$$

Discuss:

(a) the geometric meaning of each metric;

(b) the corresponding modes of convergence;

(c) why d_∞-convergence implies d_1-convergence;

(d) why the converse may fail;

(e) completeness under each metric;

(f) whether the two metrics are topologically equivalent.

A useful sequence for comparison is a family of continuous spikes with increasing height and decreasing width. The total area can tend to zero while the maximum height remains fixed.

Appendix A

Solution Sketches

A.1 Selected Week 1 Solutions

Squared distance. The function $|x-y|^2$ fails the triangle inequality: with $x = 0, y = 1, z = 2$,

$$|0-2|^2 = 4 > 1 + 1 = |0-1|^2 + |1-2|^2.$$

Disjoint neighborhoods. If $x \neq y$, let $r = d(x, y)/3$. If a point belonged to both $\mathrm{B}(x, r)$ and $\mathrm{B}(y, r)$, the triangle inequality would imply $d(x, y) < 2r < d(x, y)$, a contradiction.

A.2 Selected Week 2 Solutions

Discrete convergence. If $x_n \to x$, take $\varepsilon = 1/2$. Eventually $d(x_n, x) < 1/2$, which forces $x_n = x$. Conversely, an eventually constant sequence converges.

Equality on a dense set. For $x \in X$, take $d_n \in D$ with $d_n \to x$. Then $f(d_n) = g(d_n)$, while continuity gives $f(d_n) \to f(x)$ and $g(d_n) \to g(x)$. Uniqueness of limits gives $f(x) = g(x)$.

A.3 Selected Week 3 Solutions

Compact implies complete. Let (x_n) be Cauchy. Compactness gives a convergent subsequence $x_{n_k} \to x$. A Cauchy sequence with a convergent subsequence converges to the same limit.

Bounded but not totally bounded. Let X be an infinite set with the discrete metric. Its diameter is 1, so it is bounded. But balls of radius 1/2 are singletons, so no finite collection covers X.

A.4 Selected Week 4 Solutions

Coordinatewise convergence. For the maximum metric,

$$d_\infty((x_n, y_n), (x, y)) < \varepsilon$$

is equivalent to both coordinate distances being less than ε. Equivalence of the standard product metrics extends the result to d_1 and d_2.

Product separability. If D_X and D_Y are countable dense subsets, then $D_X \times D_Y$ is countable. Every product ball contains a point of this product, so it is dense.

Appendix B

Glossary of Core Terms

Accumulation point
A point whose every deleted neighborhood meets the set.

Boundary
Points whose every neighborhood meets both a set and its complement.

Cauchy sequence
A sequence whose terms eventually become arbitrarily close to each other.

Closed set
A set with open complement; equivalently, one containing all sequential limits of its points.

Compactness
The finite-subcover property for open covers.

Complete space
A metric space in which every Cauchy sequence converges.

Dense subset
A subset whose closure is the whole space.

Diameter
The supremum of pairwise distances within a set.

Equivalent metrics
Metrics generating the same open sets.

Interior
The largest open set contained in a given set.

Isometry
A distance-preserving map.

Metric
A distance function satisfying positivity, identity, symmetry, and the triangle inequality.

Open set
A set containing a ball around each of its points.

Product metric
A metric on a Cartesian product constructed from metrics on the factors.

Separable space
A space with a countable dense subset.

Sequential compactness
The property that every sequence has a convergent subsequence.

Total boundedness
The existence of finite ε-nets for every $\varepsilon > 0$.

Uniform continuity
Continuity with a δ that is uniform across the domain.

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